

TRUCKSON

Transportation Management System

Requirements Specification Document

Minimum Viable Product (MVP) – Version 1.0

Prepared for: Owner / Trucking Company

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1. Executive Summary

TrucksOn is a modern, web-based Transportation Management System (TMS) designed specifically for small to mid-sized trucking companies. This MVP delivers a complete operational loop from customer onboarding through load completion, invoicing, and weekly accounting — all while maintaining a clean, modular architecture that supports future growth and potential commercialization.

The system prioritizes usability on tablets (iOS and Android), cost-effectiveness, strong data security, and automated backup with ransomware protection. It is designed to run on affordable hardware (including Docker on a UGREEN NAS) while remaining scalable to hundreds or thousands of trucks.

2. Project Scope & Objectives

2.1 MVP Scope

This document defines the Minimum Viable Product (MVP). The goal is to deliver a fully functional system that a small trucking company can use daily for operations, dispatch, accounting, and basic compliance.

2.2 Key Objectives

- Provide a single source of truth for customers, drivers, trucks, trailers, loads, and maintenance.
- Enable fast dispatch creation via manual entry or AI-assisted PDF upload.
- Deliver accurate weekly accounting and basic invoicing.
- Ensure the application works cleanly on desktop and tablets.
- Implement strong security, automated backups, and ransomware protection from day one.
- Maintain a modular, Docker-based architecture that is easy to expand and eventually sell.

3. User Management & Authentication

All users must authenticate with a username and password. Role-based access control (RBAC) restricts module access based on job function.

3.1 Supported Roles (MVP)

- Admin – Full access to all modules and settings
- Dispatcher – Access to Dispatch, Load Management, Customers, Dashboard
- Driver – Limited view (future mobile app)
- Accountant – Access to Accounting, Invoicing, Reports
- Maintenance – Access to Maintenance module only

3.2 Security Requirements

- Strong password policy + future support for 2FA
- Session timeout and audit logging of critical actions
- Principle of least privilege enforced

4. Customer Management Module

Central repository for all customers and brokers. Every load must be linked to a customer.

Field	Description
Customer ID	Auto-generated unique identifier
Company Name	Legal company name
Contact Person	Primary contact name

Phone / Mobile	Primary phone number
Email	Primary email address
Billing Address	Full billing address
Payment Terms	Net 15, Net 30, COD, etc.
Notes	Free-text notes about the customer
Active Status	Active / Inactive flag

5. Driver Management Module

Field	Description
Driver ID	Auto-generated unique ID
Full Name	First + Last name
License Number	Driver's license number
License Expiration	Expiration date of license
Date of Birth	DOB for compliance
Hire Date	Date hired
Pay Per Mile Rate	Cents per mile (e.g., 0.45 = 45¢/mile)
Document Upload	License copy, medical card, other docs (multiple files)
Status	Active / Inactive / Terminated

6. Truck Management Module

Field	Description
Truck ID / Unit #	Unique unit number (e.g., TRK-101)
Make / Model / Year	Manufacturer, model name, model year
VIN	Vehicle Identification Number
In-Service Date	Date truck entered service
Out-of-Service Date	Date retired (blank = still active)
Monthly Cost	Fixed monthly ownership/lease cost
Document Upload	Registration, insurance, inspection docs
Current Status	Available / In Use / Maintenance / Retired

7. Trailer Management Module

Functionally identical to Truck Management with the addition of trailer-specific fields if needed later. Monthly cost tracking is included for accurate accounting.

Field	Description
Trailer ID / Unit #	Unique trailer number (e.g., TRL-205)
Make / Model / Year	Manufacturer and model year
In-Service / Out-of-Service	Service dates
Monthly Cost	Fixed monthly cost
Document Upload	Registration, inspection, insurance
Current Status	Available / In Use / Maintenance / Retired

8. Maintenance Module

Track all repairs and preventive maintenance for trucks and trailers.

Field	Description
Maintenance ID	Auto-generated
Equipment Type	Truck or Trailer
Equipment ID	Link to specific truck/trailer
Date Completed	Date repair/service finished
Description of Work	Detailed description of repairs performed
Cost	Total cost of repair (labor + parts)
Receipt / Invoice Upload	Attach PDF/image of receipt
Technician / Shop	Who performed the work

9. Load Management Module (Core of the System)

This is the central module. Every load has a full lifecycle with clear status tracking.

9.1 Load Status Workflow (MVP)

- Pending – Load created but not yet assigned
- Assigned – Driver, truck, and trailer assigned
- In Transit – Load picked up and en route
- Delivered – Delivered to destination
- Completed – Paperwork received and verified
- Billed – Invoice generated and sent

9.2 Key Load Fields

Field	Description
Load ID / Number	Unique load identifier (e.g., LD-2026-0847)
Customer	Link to Customer record
Status	Current status from workflow above
Pickup Location & Time	Full address + scheduled pickup datetime
Delivery Location & Time	Full address + scheduled delivery datetime
Assigned Driver	Link to Driver record
Assigned Truck	Link to Truck record
Assigned Trailer	Link to Trailer record
Rate / Revenue	Total revenue for this load
Miles (Calculated)	Distance via Google Maps API
Rate Per Mile	Auto-calculated: Revenue ÷ Miles
Notes / Activity Log	Free-text notes + timestamped activity history
Documents	BOL, POD, rate confirmation, photos, etc.

10. Dispatch Module

Create and manage dispatches. Supports both manual entry and AI-assisted PDF upload.

10.1 Core Features

- Manual creation of dispatch with truck, driver, trailer assignment
- Drag-and-drop PDF upload (rate confirmation / load tender)
- AI extraction of: Customer, Pickup/Delivery locations & times, Rate, Special terms
- Automatic distance calculation using Google Maps Directions API
- Automatic Rate Per Mile calculation ($\text{Revenue} \div \text{Miles}$)
- Link to existing Load record or create new one

10.2 Future Enhancement (Out of MVP)

- Toll cost estimation (marked for later integration)

11. Accounting Module

Weekly performance reports (Monday through Sunday) for trucks and drivers.

Report Metric	Description
Number of Loads	Completed loads in the week
Total Miles	Sum of miles driven that week
Total Revenue	Sum of revenue from completed loads
Average Revenue per Mile	$\text{Total Revenue} \div \text{Total Miles}$
Driver Pay (Calculated)	$\text{Miles} \times \text{Driver's stored pay-per-mile rate}$

12. Invoicing Module (Basic)

Generate simple invoices directly from completed loads.

- Select one or more completed loads for a customer
- Auto-populate invoice with load details, miles, rate, and total
- Basic invoice template with company logo placeholder
- Export as PDF
- Mark load status as “Billed” when invoice is generated
- Future: Email invoices directly from the system

13. Dashboard Module

Visual overview of operations with colorful charts and cards.

- Active loads currently in transit with status
- Weekly summary cards: Revenue, Miles, Loads, Avg Rate/Mile
- Available trucks and drivers count
- Upcoming or overdue maintenance alerts
- Color-coded charts (bar for revenue, pie for load status distribution)
- Quick links to create new load, dispatch, or view reports

14. Cross-Cutting Requirements

14.1 Search & Filtering

Global search across loads, customers, drivers, and trucks. Filters by date range, status, customer, driver, etc.

14.2 Notes & Activity Log

Every major record (Load, Driver, Truck, Customer) supports timestamped notes and automatic activity logging (who changed what and when).

14.3 Tablet & Mobile Responsiveness

The entire application must render cleanly and be fully usable on iOS and Android tablets (iPad, Samsung Galaxy Tab, etc.) using Chrome or Safari. No separate mobile app required for MVP.

14.4 Document Storage

All uploaded documents (licenses, receipts, BOLs, PODs, rate confirmations) are stored securely and linked to the relevant record. Quick view and download capability.

15. Technical Architecture & Deployment

15.1 Recommended Technology Stack

- Frontend: React.js (responsive, tablet-friendly)
- Backend: Python (FastAPI or Flask) – chosen for maintainability and son's familiarity
- Database: PostgreSQL (robust relational data + JSON support)
- File Storage: Local filesystem or object storage (MinIO / S3-compatible)
- PDF AI Processing: pdfplumber + cheap LLM (Groq / Mistral via OpenRouter)
- Maps: Google Maps Directions API for distance calculation
- Containerization: Docker (required)

15.2 Deployment Target

Primary production environment: UGREEN NAS (i7, 32GB RAM, 20TB storage) running Docker containers. Development occurs on a clean Linux workstation. The system is designed to be portable and can later be deployed to cloud VPS if needed for scaling or selling the software.

15.3 Why This Stack?

- Cost-effective (can run for under \$50/month at small scale)
- Docker makes backup, restore, and future scaling much easier
- Python is readable and maintainable long-term
- PostgreSQL handles complex relationships reliably

16. Backup, Security & Ransomware Protection

Data protection is a first-class requirement. The system follows the 3-2-1-1-0 backup rule with strong ransomware defenses.

16.1 Backup Strategy (Docker-based)

- Automated daily PostgreSQL database dumps (pg_dump)
- Docker volume backups for application data and uploaded documents
- Immutable snapshots on the UGREEN NAS (retention: 30 days minimum)
- Encrypted backup storage
- Offsite copy option (future: encrypted cloud storage)
- Weekly automated restore test (verification that backups are valid)

16.2 Ransomware Protection Measures

- No direct internet exposure of the NAS or database

- Access via VPN only for remote management
- Principle of least privilege for all user accounts
- Immutable backups that cannot be deleted or encrypted by ransomware
- Regular security updates on Docker host and containers
- Network segmentation (management network vs operational network)
- Audit logging of all file and database changes
- Documented incident response plan (who to call, how to restore)

17. Non-Functional Requirements

- Scalability: Architecture must support growth from 14 trucks to 1,000+ trucks without major rewrite
- Cost: Target monthly operating cost under \$50 for < 50 trucks; under \$700 for 1,000 trucks
- Usability: Clean, modern interface optimized for tablet use
- Reliability: 99.5% uptime target once deployed
- Security: All data encrypted in transit (TLS) and at rest where feasible
- Maintainability: Well-documented code with clear module boundaries

18. Future Enhancements (Out of MVP Scope)

- Fuel usage tracking via fuel card API or PDF import
- Toll cost calculation and reporting
- Full customer portal for load tracking
- Driver mobile app (iOS/Android native)
- Real-time GPS tracking integration
- ELD / Hours of Service compliance module
- Automated invoicing email + payment portal
- Advanced analytics and forecasting
- Multi-company / multi-tenant support for selling the software
- Integration with factoring companies and load boards

— End of Requirements Specification —

This document is ready for development. All core modules, data relationships, security, and deployment considerations have been defined for a complete and production-ready MVP.